

Numerical Methods

Midterm 3

June 14, 2011

- Please give details of your calculation. A direct answer without explanation is not counted.
- Your answers must be in English.
- Please carefully read problem statements.
- During the exam you are not allowed to borrow other's class note.
- Try to work on easier questions first.

Problem 1 (10%)

Given the following linear system

$$Ax = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (1)$$

1. Is the matrix A symmetric positive definite?
2. Find the solution of (1) by any method.

Problem 2 (35%)

1. Do the first CG iteration to solve (1).
2. Do the second CG iteration to solve (1).
3. Do the third CG iteration to solve (1).
4. We would like to study how the direction p_3 of the third iteration is obtained. Put your obtained r_2 , p_1 and p_2 into the following problem and solve

$$\min_p \|p - r_2\|^2$$

subject to $p \in \{Ap_1, Ap_2\}^\perp$.

Explain your findings.

Hint: the calculation may be complicated. However, you should know how to check results after each iteration. Then this task will not be that difficult.

Problem 3 (15%)

In deriving the direction p_k of conjugate gradient method, we use a lemma to know that we need to solve

$$\min_z \|r_{k-1} - AP_{k-1}z\|.$$

Then after some derivations, we have that

$$\|r_{k-1} - AP_{k-1}z\|^2 = \left(1 + \frac{\mu}{\alpha_{k-1}}\right)^2 \|r_{k-1}\|^2 + \left\| -\frac{\mu}{\alpha_{k-1}} r_{k-2} - AP_{k-2}w \right\|^2, \quad (2)$$

where

$$z = \begin{bmatrix} w \\ \mu \end{bmatrix}$$

Formally prove that if w, μ are the optimal solution of minimizing (2), then $-w/(\mu/\alpha_{k-1})$ must be the solution of

$$\min_z \|r_{k-2} - AP_{k-2}z\|.$$

Problem 4 (15%)

Given any linear system

$$Ax = b, \quad (3)$$

with A symmetric positive definite. Assume now we solve

$$P^T APx = P^T b, \quad (4)$$

instead, where P is a permutation matrix.

1. If we apply CG to solve both (3) and (4), will the number of iterations be different? We assume exact operations without any numerical errors. We also assume that the stopping condition is when the exact solution of (3) or (4) has been found.
2. How about vectors r_k, p_k and scalars α_k, β_k ? If they are changed, what are the new values?

Problem 5 (25%)

Given three points $(1, 3)$, $(2, 1)$, and $(3, 3)$. Find the spline approximation. Draw a figure to show how $s_j(x)$ looks like.

(a) Consider the following boundary condition:

$$s''_0(x_0) = 0 \quad \text{and} \quad s''_{n-1}(x_n) = 0$$

(b) Consider the following boundary condition:

$$s'_0(x_0) = -1 \quad \text{and} \quad s'_{n-1}(x_n) = 1$$